

Here are the solutions to the math questions that are posted in Blackboard.

1. For  $x \in \mathbb{R}$ , compute  $1 + x + x^2 + \cdots + x^7$ . We have

$$\begin{aligned} 1 + x + x^2 + \cdots + x^7 &= \frac{1-x}{1-x}(1 + x + x^2 + \cdots + x^7) \\ &= \frac{1}{1-x}(1 + x + x^2 + \cdots + x^7 - x - x^2 - x^3 - \cdots - x^8) \\ &= \boxed{\frac{1-x^8}{1-x}}. \end{aligned}$$

Similarly, for any positive integer  $n$  we have the general relation:

$$\sum_{j=0}^n x^j = 1 + x + \cdots + x^n = \frac{1-x^{n+1}}{1-x}. \quad (1)$$

2. For  $0 < x < 1$  compute

- (a)  $1 + x + x^2 + \cdots$ . To find this sum just take the limit of (1) as  $n$  approaches infinity to get

$$\sum_{j=0}^{\infty} x^j = \lim_{n \rightarrow \infty} \sum_{j=0}^n x^j = \lim_{n \rightarrow \infty} \frac{1-x^{n+1}}{1-x} = \boxed{\frac{1}{1-x}}, \quad (2)$$

since  $|x| < 1$ .

- (b)  $1 + 2x + 3x^2 + \cdots$ . This is very similar to question (a) if you are careful to notice the following:

$$1 + 2x + 3x^2 + \cdots = \sum_{j=1}^{\infty} jx^{j-1} = \sum_{j=1}^{\infty} \left( \frac{d}{dx} x^j \right).$$

At this point we cross our fingers and we exchange the order of the derivative with the infinite sum<sup>1</sup> to get

$$\begin{aligned} \sum_{j=1}^{\infty} \left( \frac{d}{dx} x^j \right) &= \frac{d}{dx} \left( \sum_{j=1}^{\infty} x^j \right) = \frac{d}{dx} \left[ \left( \sum_{j=0}^{\infty} x^j \right) - 1 \right] \stackrel{(2)}{=} \frac{d}{dx} \left[ \left( \frac{1}{1-x} \right) - 1 \right] \\ &= \boxed{\frac{1}{(1-x)^2}}. \end{aligned}$$

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<sup>1</sup>... because, yes, it's not always allowed when we have an infinite sum, but in this case it is!

(c)  $x + \frac{x^2}{2} + \frac{x^3}{3} + \dots$ . Same idea as question (b):

$$\begin{aligned} x + \frac{x^2}{2} + \frac{x^3}{3} + \dots &= \sum_{j=0}^{\infty} \frac{x^{j+1}}{j+1} = \sum_{j=0}^{\infty} \left( \int_0^x t^j dt \right) = \int_0^x \left( \sum_{j=0}^{\infty} t^j \right) dt \\ &\stackrel{(2)}{=} \int_0^x \frac{1}{1-t} dt = \int_0^x \frac{d}{dt} (-\log(1-t)) dt \\ &= \boxed{-\log(1-x)}. \end{aligned}$$

3.  $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ . This one is taken **by definition** as

$$\boxed{e^x} := \sum_{j=0}^{\infty} \frac{x^j}{j!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots.$$

In fact, this is true for every  $x$  in  $\mathbb{R}$ . It is also not hard to check that the series is exactly the Taylor series expansion of  $e^x$ .

4. It is known that for every number  $\lambda > 0$  and every positive integer  $k$

$$\int_0^{\infty} \frac{\lambda e^{-\lambda x} (\lambda x)^k}{k!} dx = 1. \quad (3)$$

Then to calculate the following integrals we always try to write their integrands in the same format as the integrand of (3).

(a) To that end,

$$\int_0^{\infty} e^{-2x} x^5 dx = \frac{5!}{2 \cdot 2^5} \underbrace{\int_0^{\infty} \frac{2e^{-2x} (2x)^5}{5!} dx}_1 = \frac{120}{64} = \boxed{\frac{15}{8}}.$$

(b) Also,

$$\int_0^{\infty} x^2 \frac{e^{-x} x^5}{5!} dx = \frac{1}{5!} \int_0^{\infty} e^{-x} x^7 dx = \frac{7!}{5!} \underbrace{\int_0^{\infty} \frac{e^{-x} x^7}{7!} dx}_1 = 7 \cdot 6 = \boxed{42}.$$